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Fault detection for DC-DC converters using adaptive parameter identification[☆]

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Abstract

As a clean and environmentally friendly renewable energy source, photovoltaic (PV) energy has attracted widespread attention. DC-DC converters are an essential component of all PV systems and it is vital for them to maintain normal function. Therefore, research on fault detection and identification in DC-DC converters is significant. This paper designs a new type of adaptive sliding mode observer, which realizes parameter estimation and fault detection of components by ensuring the consistent boundedness of parameter estimation errors. When the system fails, the parameter estimated value and actual value of the observer will generate a residual signal which can be analyzed and compared with the designed threshold value. If the threshold value is exceeded, the component will be diagnosed as malfunctioning. Finally, the effectiveness of the method is verified by a digital online simulation and hardware-in-the-loop simulation.

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1. Introduction

Rapid economic development has caused the problem of environmental destruction while non-renewable energy depletion is becoming more and more obvious. Photovoltaic (PV) power

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generation technology is perceived as a friendly means for the environment and has many applications. In addition, as the proportion of solar power in the global grid increases every year, the possibility of system failure also increases. PV systems' stable and safe operation is thus essential to the power grid and consumers [1–4]. DC-DC converters are an important element of PV systems [5]. If changes to the parameters of DC-DC converter components are not detected in time, they can easily damage the converter, which will eventually cause the PV system to collapse. The fault types of DC-DC converters can be divided into switch fault [6–8], component fault [9,10] and sensor fault [11–13]. In the previous literature, research on DC-DC converter faults mainly focuses on switching faults, and there is little research on component parameter faults.

According to the investigation, converter failure is mainly related to the capacitor and the power switch. A large body of literature exists on the failure of DC-DC converters [14]. In [15], a fault diagnosis method for a boost converter for a PVMPPT system is proposed. Jain et al. [16] improves the state observer to address the interference problem of noise and interference. Based on the switching system model of switching power converters, literature [17] proposes a method to detect and isolate different types of faults. Silveira and Araújo [18] proposes a new solution to the problem of different types of faults in DC-DC power converters, which can effectively detect and isolate faults. The attack detection method can also be applied to DC-DC converters. Li et al. [19], Su et al. [20] use a sliding mode observer to estimate the system state, generate residuals to determine whether the system has a fault, and then estimate the fault value through an adaptive sliding mode observer. The disadvantage of this technique for fault detection and separation based on generated residuals is that it cannot directly estimate faults. In [21], a comprehensive fault detection and aging monitoring system is proposed for DC-DC converters, which can realize the fault detection of switch faults, diode faults and inductance-to-turn faults and the aging monitoring of capacitor switches. However, this method is insufficient for real-time fault diagnosis. In [22], the generalized gradient descent method is used to realize the parameter identification of the components of the interleaved DC-DC converter. Cheng et al. [23,24], Tao et al. [25], Casavola and Gagliardi [26] describe disturbances, modeling errors, the effects of various uncertainties, and robust filtering techniques in natural systems. [27–29] propose fault detection and isolation based on LPV observer, fault-tolerant sensor coordination scheme and LPV modeling for fault detection and isolation. This kind of approach can be useful when designing FDI strategies for complex nonlinear systems due to the approximation properties of the LPV (qLPV) models. The choice of the LPV framework can be mainly justified by the fact that it can compactly describe plants subject to transitions among different operating conditions related to the variations of some fundamental system parameters. These may account for the physical modifications of the plant itself and model switching due to linearization at different operating points [30].

An FDI system is advantageous because it can contribute to the faulty tolerance with minimal additional cost and system complexity (instead of an approach using component redundancy). An FDI system carries out two tasks: (1) detection, which involves making a binary decision on whether or not a fault has occurred, and (2) identification, which determines the location of the faulty component. It is important to note that an FDI system, does not in general replace hardware-based fault protection devices, such as fuses, circuit breakers, and internal semiconductor protection modules. Instead, an FDI system provides online granular information about the converter's condition and health, which can be used for monitoring, prognosis, and automatic fault remediation.

While these two methods can better estimate component parameters in real time, the fault detection method proposed in this paper can respond to faults faster and has better results.

This paper studies the problem of component fault in boost converters in photovoltaic production. The observer can detect faults, identify fault locations and estimate fault parameters through an adaptive sliding mode observer designed for DC-DC converters. The observer can quickly track the system parameters when the system is fault-free. Once the system fails, the observer can identify the fault parameter value, and if the residual signal exceeds the threshold, the system will set off an alarm. To verify the effectiveness of the method, we verified its feasibility by simulating the boost converter of a solar photovoltaic system. At the same time, the power electronics hardware-in-the-loop simulator Typhoon-HIL402 and the digital controller DSP28335 were used for hardware-in-the-loop simulation. Regarding verifying the effectiveness of the method, the first challenge is to ensure the accuracy of the adaptive law when estimating device parameters online. This paper derives the appropriate sliding mode signal and adaptive law based on the designed adaptive sliding mode observer. It uses the Lyapunov function to analyze the boundedness of the state estimation error and parameter estimation error to overcome the difficulties and challenges mentioned above. Another challenge is determining the specific location of the fault when the system fails. In this paper, a threshold analysis system is designed to detect the fault and determine its specific location. We compared the proposed method with the common residual error generating way by the observer's estimated value and the common value.

The rest of this paper is organized as follows. In [Section 2](#), the mathematical model of the converter used to perform this study is presented. In [Section 3](#), an adaptive sliding mode observer is designed for fault diagnosis and the fault identification scheme is given. The simulation results are presented in [Section 4](#). In [Section 5](#), it introduces the semi-physical simulation experiment and results. A summary of this article is given at the end.

2. Mathematical model

2.1. Boost converters model

The analyzed DC-DC Boost converter circuit is shown in [Fig. 1](#). The Boost converter has two switching modes: in mode 1, SW_1 is closed and SW_2 is open; in mode 2, SW_1 is open and

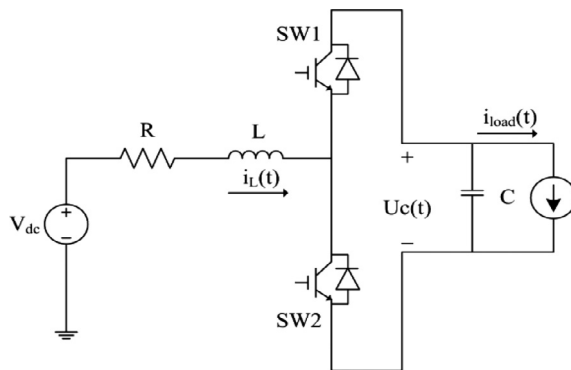


Fig. 1. DC-DC Boost converter.

SW_2 is closed. SW_1 and SW_2 are two insulated-gate transistor-diode switches. Selecting the capacitor voltage $U_c(t)$ and the inductor current $i_L(t)$ as the state of the system, respectively expressed by variables $x_1(t)$ and $x_2(t)$. The input voltage $V_{dc}(t)$ and the load current $i_{load}(t)$ are used as the input of the system, which are represented by $u_1(t)$ and, respectively. The following model can be obtained:

In mode 1 (SW_1 is closed at the same time SW_2 is opened), it has

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{L} & 0 \\ 0 & -\frac{1}{C} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

In mode2 (SW_1 is opened at the same time SW_2 is closed), it has

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{L} & 0 \\ 0 & -\frac{1}{C} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

The circuit switching system model is expressed as follows:

$$\begin{aligned} \dot{x}(t) &= A_{\sigma(t)}x(t) + B_{\sigma(t)}u(t) \\ y(t) &= Cx(t) \end{aligned} \quad (1)$$

where $x(t) = [i_L(t) \quad U_c(t)]^T$ is the state of the switching system, $y(t)$ is the output vector of the system, and $u(t) = [V_{dc}(t) \quad i_{load}(t)]^T$ is the external input, $A_{\sigma(t)}$, $B_{\sigma(t)}$, C is a collection of state space models,

$$A_{\sigma(t)} = \begin{bmatrix} -\frac{R}{L} & -\frac{s_1}{L} \\ \frac{s_1}{C} & 0 \end{bmatrix}, B_{\sigma(t)} = \begin{bmatrix} \frac{1}{L} & 0 \\ 0 & -\frac{1}{C} \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

the switching signal $\sigma(t)$ is controlled by the switch state. Considering the switches $SW_1 \sim SW_2$ in Fig. 1, the switching state of the DC boost converter is as follows:

$$s_1 = \begin{cases} 1, & SW_1 \text{ is closed at the same time } SW_2 \text{ is opened} \\ 0, & SW_1 \text{ is opened at the same time } SW_2 \text{ is closed} \end{cases}$$

2.2. Fault description

Taking the Boost converter as an example, consider the component fault in the DC-DC Boost converter. The components' fault model is described as follow:

$$\begin{aligned} \dot{x}(t) &= A_{\sigma(t)}x(t) + B_{\sigma(t)}u(t) + Hf_i(t) \\ y(t) &= Cx(t) \end{aligned} \quad (2)$$

where $H = I$ with appropriate dimensions, $i = 1, 2 \dots w$, w is the number of fault components. $f_i(t)$ is the fault signal generated by the component.

For the fault signal, the fault model is as follows:

$$f_i(t) = \delta_i \varphi_n(t) \quad (3)$$

Denote δ_i is the component fault parameter. $\varphi_n(t)$ is the fault component. (n indicates fault type)

Example 1. Consider a fault in the inductance L that the value of the inductance L changes to $L + \Delta L$, where ΔL describes the magnitude of the fault. Thus, the system dynamics can be described as

$$\dot{x}(t) = \bar{A}_{\sigma(t)}x(t) + \bar{B}_{\sigma(t)}u(t) \quad (4)$$

where

$$\bar{A}_{\sigma(t)} = \begin{bmatrix} -\frac{R}{L+\Delta L} & -\frac{s_1}{L+\Delta L} \\ \frac{s_1}{C} & 0 \end{bmatrix}, \bar{B}_{\sigma(t)} = \begin{bmatrix} \frac{1}{L+\Delta L} & 0 \\ 0 & -\frac{1}{C} \end{bmatrix} \quad (5)$$

Then system Eq. (4) can be written as

$$\begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = \begin{pmatrix} -R/(L+\Delta L) & -s_1/(L+\Delta L) \\ s_1/C & 0 \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} + \begin{pmatrix} 1/(L+\Delta L) & 0 \\ 0 & -1/C \end{pmatrix} \begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix}$$

After transformation, the fault matrix is obtained

$$f_1(t) = \left(\frac{1}{L+\Delta L} (-Ri_L - s_1U_c + V_{dc}) \quad 0 \right)^T$$

Fault component $\varphi_1(t) = [(-Ri_L(t) - s_1U_c(t) + V_{dc}) \quad 0]^T$, component fault parameter $\delta_1 = \frac{1}{L+\Delta L}$. Similarly, capacitance fault $f_2(t)$ can be concluded as follows

$$f_2(t) = \left(0 \quad \frac{1}{C+\Delta C} (s_1i_L(t) - i_{load}(t)) \right)^T$$

Fault component $\varphi_2(t) = [0 \quad (s_1i_L(t) - i_{load}(t))]^T$, component fault parameter $\delta_2 = \frac{1}{C+\Delta C}$

Several fault conditions of simulated Boost converter circuit component failure can be represented by the following table:

According to the established fault model, an adaptive sliding mode observer is proposed, which enables the component faults to be reconstructed online by the adaptive law. Before presenting the theorem, an assumption is given.

Assumption 1. There is a constant ρ such that $\|\varphi_n\| \leq \rho$.

3. Fault detection

In this part, for the component failure of the DC-DC converter, the adaptive sliding mode observer is designed to estimate the parameter value of the circuit in real time, and the Lyapunov function proves the uniform boundedness of the parameter estimation error. The parameter estimated value and the actual value of the detector form a residual signal, and the fault detection is realized by comparing the residual signal with the threshold value.

3.1. Adaptive sliding mode observer design

Adaptive sliding-mode observer possesses both the advantages of the adaptive method and the sliding-mode method. The sliding-mode variable structure strategy has good robustness and insensitiveness to interference, but it also exists the chattering phenomenon and the high-frequency oscillation. In the other hand, the adaptive strategy is good at estimating parameters and tracking objectives. Therefore, adaptive sliding mode observer is used to realize fault diagnosis of components.

Design an adaptive sliding mode observer with the structure shown in Eq. (6)

$$\begin{aligned} \dot{\hat{x}}_i(t) &= A_{\sigma(t)}\hat{x}_i(t) + B_{\sigma(t)}u(t) + G_i(y(t) - \hat{y}_i(t)) + H v_i(t) \\ \hat{y}_i(t) &= C\hat{x}_i(t) \\ r_i(t) &= y(t) - \hat{y}_i(t) \end{aligned} \quad (6)$$

where $\hat{x}_i(t)$ is the state estimation vector, G_i is the observer gain matrix. $i = 1, 2 \dots w$, w is the number of fault types.

Define weight matrix $\hat{f}_i(t) = M_{f_i}(s)f_i(t)$, among them $M_{f_i}(s)$ is a custom matrix, at this time $M_{f_i}(s)$ the minimum realization of is:

$$\begin{aligned}\dot{x}_{m_i}(t) &= A_m x_{m_i}(t) + B_m f_i(t) \\ \hat{f}_i(t) &= C_m x_{m_i}(t) + D_m f_i(t)\end{aligned}\quad (7)$$

Among them, A_m, B_m, C_m, D_m are constant matrices, $x_{m_i}(t)$ is the state vector of the weighted system, and $\hat{f}_i(t)$ is the weighted fault signal.

Define the residual $e_i(t) = x(t) - \hat{x}_i(t)$, $r_{e_i}(t) = r_i(t) - \hat{f}_i(t)$, $\tilde{x}(t) = [e_i^T(t) \ x_{m_i}^T(t)]$, $e_{y_i}(t) = y(t) - \hat{y}_i(t)$ then the weighted system matrix can be rewritten as:

$$\begin{aligned}\dot{\tilde{x}}_i(t) &= \begin{bmatrix} A_{\sigma(t)} - G_i C & 0 \\ 0 & A_m \end{bmatrix} \tilde{x}_i(t) + \begin{bmatrix} H \\ B_m \end{bmatrix} f_i(t) - \begin{bmatrix} H \\ 0 \end{bmatrix} v_i(t) \\ r_{e_i}(t) &= [C \quad C_m] \tilde{x}(t) + [-D_m] f_i(t)\end{aligned}\quad (8)$$

Consider the switching system Eq. (8) to obtain the observer gain under any switching signal condition, and the switching system Eq. (8) satisfies the following two requirements:

- (1) The switching system Eq. (8) is asymptotically stable.
- (2) Under zero initial state and arbitrary switching signal, the fault detection observer satisfies the H_∞ performance index of formula Eq. (9), which minimizes the influence of external fault signals on the residual signal $r_{e_i}(t)$ and enhances the system's performance anti-interference can be improved.

$$\int_0^\infty r_{e_i}^T(t) r_{e_i}(t) dt < \gamma^2 \int_0^\infty f_i^T(t) f_i(t) dt \quad (9)$$

In order to obtain the gain of the fault detection observer and meet the performance index Eq. (9), we give the following lemma to realize the design of the fault detection observer.

Lemma 1 ([19]). Given constant $a_{1\sigma_i}, a_{2\sigma_i}, a_{3\sigma_i}$, if it exists $W_{1_i}, W_{2\sigma_i}, W_{3\sigma_i}, W_{4\sigma_i}, W_{5\sigma_i}, L_i$, the positive definite symmetric matrix P makes the following inequality true:

$$\begin{bmatrix} \phi_{11} & \phi_{12} & \phi_{13} & 0 \\ * & \phi_{22} & \phi_{23} & \tilde{C}^T \\ * & * & -\gamma^2 H & \tilde{D}^T \\ * & * & * & -H \end{bmatrix} < 0 \quad (10)$$

where:

$$\begin{aligned}\phi_{11} &= \begin{bmatrix} -He(W_{1_i}) & -a_{1\sigma} W_{1_i} - W_{2\sigma_i}^T \\ * & -He(W_{3\sigma_i}) \end{bmatrix}, \\ \phi_{12} &= \begin{bmatrix} P_1 - a_{2\sigma} W_{1_i} + W_{1_i}^T A_\sigma - L_i C & P_2 - a_{3\sigma} W_{1_i} + W_{2\sigma_i}^T A_w \\ P_2^T - W_{4\sigma_i} + a_{1\sigma} W_{1_i}^T A_\sigma - a_{1\sigma} L_i C & P_3 - W_{5\sigma_i} + W_{3\sigma_i}^T A_w \end{bmatrix}, \\ \phi_{13} &= \begin{bmatrix} W_{1_i}^T + W_{2\sigma_i}^T B_w \\ a_{1\sigma} W_{1_i}^T + W_{3\sigma_i}^T B_w \end{bmatrix},\end{aligned}$$

$$\begin{aligned}\phi_{23} &= \begin{bmatrix} a_{2\sigma}W_{1_i}^T + W_{4\sigma_i}^TB_w \\ a_{3\sigma}W_{1_i}^T + W_{5\sigma_i}^TB_w \end{bmatrix}, \\ \phi_{22} &= \begin{bmatrix} a_{2\sigma}He(W_{1_i}^TA_\sigma - L_iC) & W_{4\sigma_i}^TA_w + a_{3\sigma}A_\sigma^TW_{1_i} - a_{3\sigma}C^TL_i^T \\ * & He(W_{5\sigma_i}^TA_w) \end{bmatrix}\end{aligned}$$

Then switching system Eq. (8) is asymptotically stable and can guarantee the H_∞ performance index of formula Eq. (9). The observer gain matrix is obtained by the formula $G_i = (W_{1_i}^T)^{-1}L_i$.

$v_i(t)$ is the sliding mode variable structure input signal, and its expression is shown in Eq. (11).

$$v_i(t) = k_1 \hat{\delta}_i(t) \rho \frac{Pe_i(t)}{\|Pe_i(t)\| + \tau} \quad (11)$$

where k_1 is the design parameter and τ is an arbitrarily small positive real number to ensure the continuity of the nonlinear term. ρ is the constant given by Assumption 1. δ_i is component fault parameter. $\hat{\delta}_i(t)$ is an estimate of δ_i given in Eq. (11). Adaptive law which is as follows:

$$\dot{\hat{\delta}}_i(t) = -a_1 \hat{\delta}_i(t) + a_2 \rho \frac{\|Pe_i(t)\|^2}{\|Pe_i(t)\| + \tau} \quad (12)$$

where a_1 and a_2 are design parameters.

Remark 1. In most sliding mode observer designs, the nonlinear term $v_i(t)$ is discontinuous. Therefore, the introduction of continuous small quantity is considered here to ensure the continuity of the nonlinear term. At the same time, the adaptive law is proposed to estimate the fault parameter values for unknown faults.

The dynamic characteristics of the estimation error can be described as follows:

$$\dot{e}_i(t) = (A_{\sigma(t)} - G_i C_{\sigma(t)})e_i(t) + H(f_i(t) - v_i(t)) \quad (13)$$

In order to achieve the accuracy of signal estimation of component parameters, it is necessary to ensure that the state estimation error of the sliding mode observer reaches the sliding mode surface in a infinite time, and the parameter estimation error should be as small as possible. This section will prove the stability of the adaptive sliding mode observer. Considering the system of Eq. (1), the following argument holds when the conditions of Assumption 1 is satisfied.

Theorem 1. Under Assumption 1, the state estimation error $e_i(t)$ and the parameter estimation error $\tilde{\delta}_i(t) = \delta_i - k_1 \tilde{\delta}_i(t)$ are both bounded, and they will converge to the following set at exponential speed.

$$\phi = \left\{ (e_i(t), \tilde{\delta}_i(t)) \mid \|e_i(t)\| < \sqrt{\frac{2\mu_1}{p_n\mu_0}}, |\tilde{\delta}_i(t)| < \sqrt{\frac{2k_1a_2\mu_1}{\mu_0}} \right\} \quad (14)$$

Denote $\mu_1 = \frac{a_1}{k_1a_2}\delta_i^2 + \delta_i\rho\tau$, $\mu_0 = \min(\frac{q_n}{p_m}, 2a_1)$, $p_n = \lambda_{\min}(P)$, $p_m = \lambda_{\max}(P)$ and $q_n = \lambda_{\min}(Q)$.

Proof. We choose the common Lyapunov function $V(t) = V_1(t) + V_2(t)$, denote $V_1(t) = \frac{1}{2}e_i^T(t)Pe_i(t)$ and $V_2(t) = \frac{1}{2k_1a_2}\tilde{\delta}_i^2(t)$. Then the derivative of $V(t)$ can be described as fol-

lows:

$$\dot{V}_1(t) = M_1 + M_2 \quad (15)$$

where

$$M_1 = \frac{1}{2} e_i^T(t) [(A_{\sigma(t)} - G_i C_{\sigma(t)})^T P + P(A_{\sigma(t)} - G_i C_{\sigma(t)})] e_i(t) \quad (16)$$

$$M_2 = \frac{1}{2} [f_i(t) - v_i(t)]^T H^T P e_i(t) + \frac{1}{2} e_i^T(t) P H [f_i(t) - v_i(t)] \quad (17)$$

The following will analyze the characteristics of M_1 and M_2 . we get M_1

$$M_1 < -\frac{1}{2} e_i^T(t) Q e_i(t) \quad (18)$$

For M_2 , it has

$$\begin{aligned} M_2 &\leq (f_i(t) - v_i(t)) \|H\| \|P e_i(t)\| \leq f_i(t) \|P e_i(t)\| - v_i(t) \|H\| \|P e_i(t)\| \\ &\leq \delta_i \varphi_n(t) \|P e_i(t)\| - k_1 \hat{\delta}_i(t) \rho \frac{\|P e_i(t)\|^2}{\|P e_i(t)\| + \tau} \end{aligned} \quad (19)$$

Since H is an identity matrix, $\|H\| = 1$, τ is an arbitrarily small real number, then $\frac{\|P e_i(t)\|}{\|P e_i(t)\| + \tau} \leq 1$. According to

[Assumption 1](#), we can obtain

$$\begin{aligned} M_2 &\leq \rho \frac{\|P e_i(t)\|}{\|P e_i(t)\| + \tau} (\delta_i (\|P e_i(t)\| + \tau) - k_1 \hat{\delta}_i(t) \|P e_i(t)\|) \\ &\leq \tilde{\delta}_i(t) \rho \|P e_i(t)\|^2 \frac{1}{\|P e_i(t)\| + \tau} + \delta_i \rho \tau \end{aligned} \quad (20)$$

Substituting [Eqs. \(20\) and \(18\)](#) into [Eq. \(15\)](#), it has

$$\dot{V}_1(t) < -\frac{1}{2} e_i^T(t) Q e_i(t) + \tilde{\delta}_i(t) \rho \frac{\|P e_i(t)\|^2}{\|P e_i(t)\| + \tau} + \delta_i \rho \tau \quad (21)$$

Due to δ_i is an unknown constant, then $\dot{\delta}_i = 0$. According to parameter estimation error $\tilde{\delta}_i(t) = \delta_i - k_1 \hat{\delta}_i(t)$, then the derivative of $V_2(t)$ can be expressed as follows

$$\dot{V}_2(t) = -\frac{\dot{\hat{\delta}}_i(t) \tilde{\delta}_i(t)}{a_2} \quad (22)$$

The following equation can be obtained by adding [Eqs. \(21\) and \(22\)](#).

$$\dot{V}(t) < -\frac{1}{2} e_i^T(t) Q e_i(t) + \tilde{\delta}_i(t) \left[\frac{\rho \|P e_i(t)\|^2}{\|P e_i(t)\| + \tau} - \frac{1}{a_2} \dot{\hat{\delta}}_i(t) \right] + \delta_i \rho \tau \quad (23)$$

Bring the adaptive law to [Eq. \(23\)](#), we can get

$$\dot{V}(t) < -\frac{1}{2} e_i^T(t) Q e_i(t) + \tilde{\delta}_i(t) \frac{a_1}{a_2} \hat{\delta}_i(t) + \delta_i \rho \tau \quad (24)$$

Simplify the above formula. The second item on the right of [Eq. \(19\)](#) can be rewritten as follows

$$\tilde{\delta}_i(t) \frac{a_1}{a_2} \hat{\delta}_i(t) = \tilde{\delta}_i(t) \frac{a_1}{k_1 a_2} \left[(\delta_i - \tilde{\delta}_i(t)) \right] \leq -\frac{a_1}{k_1 a_2} \tilde{\delta}_i^2(t) + \frac{a_1}{k_1 a_2} \delta_i^2 \quad (25)$$

Substitute Eq. (25) into Eq. (24), and we know that the third and fourth terms are constants. Next, we will analyze the first and second terms. It notes that $q_n \|e_i(t)\|^2 \leq e_i^T(t) Q e_i(t)$, $e_i^T P e_i(t) \leq p_m \|e_i(t)\|^2$. Then it has

$$\begin{aligned} -\frac{1}{2} e_i^T(t) Q e_i(t) &\leq -\frac{q_n}{p_m} V_1(t) \\ -\frac{a_1}{k_1 a_2} \tilde{\delta}_i^2(t) &= -2a_1 V_2(t) \end{aligned} \quad (26)$$

Substitute the above formula into Eq. (24), we have

$$\dot{V}(t) \leq -\frac{q_n}{p_m} V_1 - 2a_1 V_2 + \frac{a_1}{k_1 a_2} \delta_i^2 + \delta_i \rho \tau < -\mu_0 V(t) + \mu_1 \quad (27)$$

where $\mu_0 = \min(\frac{q_n}{p_m}, 2a_1) \mu_1 = \frac{a_1}{k_1 a_2} \delta_i^2 + \rho \delta_i \tau$

Multiply both sides of the above equation by $e_i^{\mu_0 t}$, it has

$$\dot{V}(t) e_i^{\mu_0 t} + e_i^{\mu_0 t} \mu_0 V(t) < \mu_1 e_i^{\mu_0 t} \quad (28)$$

Integrate on both sides of the above equation, it has

$$V(t) e_i^{\mu_0 t} - V(0) < \frac{\mu_1}{\mu_2} e_i^{\mu_0 t} \quad (29)$$

$$V(t) < V(0) e_i^{-\mu_0 t} + \frac{\mu_1}{\mu_0} \quad (30)$$

Due to $V(t) = V_1(t) + V_2(t)$ and $V_1(t) \leq \frac{1}{2} p_m \|e_i(t)\|^2$, it indicate that both $V_1(t)$ and $V_2(t)$ are less than the right side of $V_1(t) \leq \frac{1}{2} p_m \|e_i(t)\|^2$ Eq. (30).

Substitute $V_1(t)$ and $V_2(t)$ into Eq. (30), and we can get

$$\frac{1}{2} p_m \|e_i(t)\|^2 \leq V(0) e_i^{-\mu_0 t} + \frac{\mu_1}{\mu_0} \quad (31)$$

$$\frac{1}{2k_1 a_2} \tilde{\delta}_i^2(t) \leq V(0) e_i^{-\mu_0 t} + \frac{\mu_1}{\mu_0} \quad (32)$$

Using the above equation, $e_i(t)$ is bounded,

$$\|e_i(t)\|^2 \leq \frac{2}{p_n} e_i^{-\mu_0 t} V(0) + \frac{2\mu_1}{p_n \mu_0} \quad (33)$$

Similarly, $\tilde{\delta}_i^2(t)$ is bounded

$$\tilde{\delta}_i^2(t) < 2k_1 a_2 e_i^{-\mu_0 t} V(0) + 2k_1 a_2 \frac{\mu_1}{\mu_0} \quad (34)$$

Evidently, due to $\mu_0 > 0$ and $V(0)$ is constant, if $t \rightarrow \infty$, then $e_i^{-\mu_0 t} V(0) \rightarrow 0$. Then, for arbitrary p_n, k_1 and a_2 , $e_i(t)$ and $\tilde{\delta}_i(t)$ will converge to the set ϕ at exponential speed. $\square$

Theorem 2. By choosing sufficiently small a_1 and τ , and sufficiently large k_1 , the set ϕ can be arbitrarily small as defined.

Proof. Denote $\mu_0 = \min(\frac{q_n}{p_m}, 2a_1)$. For any given a_1 and k_1 , when they satisfy $2a_1 \geq \frac{q_n}{p_m}$ and $\mu_0 = \frac{q_n}{p_m}$, we get

$$\frac{\mu_1}{\mu_0 p_n} = \frac{\delta_i^2}{p_n \mu_0 k_1 a_2} + \frac{\delta_i \rho}{p_n \mu_0} \tau$$

$$k_1 a_2 \frac{\mu_1}{\mu_0} = \frac{\delta_i^2}{\mu_0} a_1 + \frac{k_1 a_2 \delta_i \rho}{\mu_0} \tau \quad (35)$$

Obviously, when choosing sufficiently large k_1 and sufficiently small a_1 and τ , both $2 \frac{\mu_1}{\mu_0 p_n}$ and $2k_1 a_2 \frac{\mu_1}{\mu_0}$ can become arbitrarily small. Thus, the set ϕ can be arbitrarily small as defined. $\square$

Remark 2. Assumption 1 is the prerequisites for realizing the stability of the adaptive sliding mode observer. Only when Assumption 1 is satisfied, the design adjustment law can reconstruct the value of the faulty component online. But Assumption 1 cannot meet the requirements of all systems, so this method has limitations.

Remark 3. Through the proof of Theorem 2, we know that the set ϕ can be arbitrarily small by definition. When k_1 becomes larger $|\delta_i(t)|$ can be arbitrarily small. When choosing a sufficiently large k_1 and a sufficiently small τ , $k_1 \hat{\delta}_i(t)$ can approximate δ_i with a sufficiently small error. The adaptive law can better reconstruct component failure values.

3.2. Fault detection logic

In the DC-DC converter fault diagnosis process, the observer's estimated value of the components and the system measurement value will produce residuals. Under healthy conditions, the residual error is very small and close to zero. When a component fails, its corresponding residual component will undergo a sudden change. Therefore, we use residual components to find specific faulty components. A threshold analysis system is established and compared with the residual value to locate the fault components, so as to realize fault detection and identification. If the threshold is chosen too large, it will cause a high false alarm rate; otherwise, it will increase the false alarm rate if the threshold is chosen too small.

Fault detection is accomplished by comparing the residual estimation function $J(r_i(t))$ with the threshold T . The mathematical model of the ideal system is the residual estimation function $J(r_i(t))$ is calculated as follows:

$$J(r_i(t)) = \sqrt{\frac{1}{M} \int_0^M r_i(t)^T r_i(t) d\tau}$$

where 0 is the initial evaluation time, M is the evaluation time. Once the evaluation function is selected, we can determine the threshold. Threshold selection in this paper is based on the upper bound of residual signal when the circuit is fault-free, which is determined as:

$$T = \sup_{f_i(t)=0} J(r_i(t)) \quad (36)$$

Fault detection can be realized by comparing residual function and threshold, taking the inductance fault as an example, the fault identification logic can be designed as follows and the detection logic is as follows:

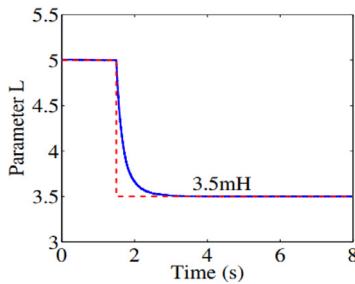
$$Flag = \begin{cases} 1, & J(r_i(t)) \geq T, \text{ fault} \\ 0, & J(r_i(t)) < T, \text{ no fault} \end{cases}$$

4. Simulation results and analysis

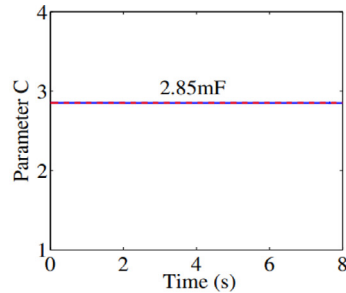
Consider the DC-DC Boost converter with component fault as shown in Fig. 1, whose parameters are $V_{dc} = 190$ V, $i_{load} = 3$ A, $R = 0.082 \Omega$, $L = 5$ mH $\pm 10\%$ and $C = 2.85$ mF $\pm$

Table 1
Boost converter circuit fault signals.

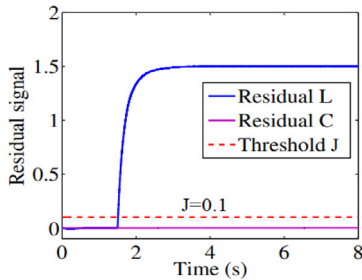
Flag type	Parameter	Fault events	Fault component
$n = 1$	$\delta_1 = \frac{1}{L}$	$\frac{1}{L} \rightarrow \frac{1}{L+\Delta L}$	$\varphi_1(t) = [(-Ri_L(t) - s_1U_c(t) + V_{dc}) \quad 0]^T$
$n = 2$	$\delta_2 = \frac{1}{C}$	$\frac{1}{C} \rightarrow \frac{1}{C+\Delta C}$	$\varphi_2(t) = [0 \quad (s_1i_L(t) - i_{load}(t))]^T$
$n = 3$	$\delta_1 = \frac{1}{L}$ $\delta_2 = \frac{1}{C}$	$\frac{1}{L} \rightarrow \frac{1}{L+\Delta L}$ $\frac{1}{C} \rightarrow \frac{1}{C+\Delta C}$	$\varphi_3(t) = \begin{bmatrix} (-Ri_L(t) - s_1U_c(t) + V_{dc}) \\ (s_1i_L(t) - i_{load}(t)) \end{bmatrix}$



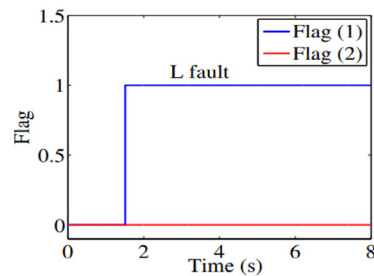
(a) Estimated value of inductance L parameters



(b) Estimated value of capacitance C parameters



(c) L and C residual diagram



(d) Fault detection flag

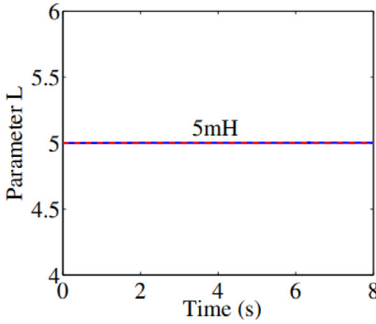
Fig. 2. Simulation of inductance L fault.

20%. The Boost converter fault types and fault signals are shown in Table 1. Then, two adaptive sliding mode observers are shown in Eq. (6) are designed, and $i \in \{1, 2\}$.

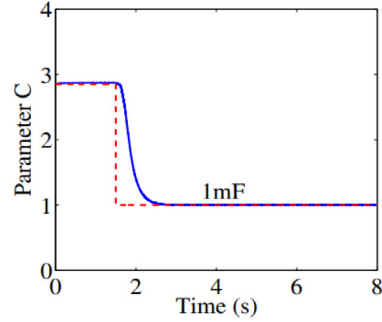
For the convenience of simulation, given $a_{11i} = 1$, $a_{12i} = 20$, $a_{21i} = 30$, $a_{22i} = 60$, $a_{31i} = -0.1$, $a_{32i} = -20$, $i \in \{1, 2\}$, then G_i and P are obtained by solving the linear matrix inequality Eq. (10).

$$G_i = \begin{bmatrix} -15.9781 & 0.0291 \\ -0.0503 & 0.7402 \end{bmatrix}, \quad P = \begin{bmatrix} 174.4447 & 0.0000 \\ 0.0000 & 99.4338 \end{bmatrix}, \quad i \in \{1, 2\}$$

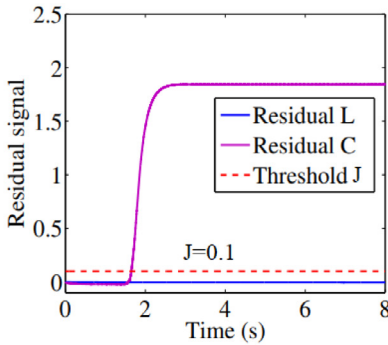
Considered the switched characteristics and the uncertainty by the model, the threshold is used as 0.1. In the following part, the effectiveness of the fault detection method is verified by MATLAB simulation and hardware-in-the-loop simulation for the failure of three components.



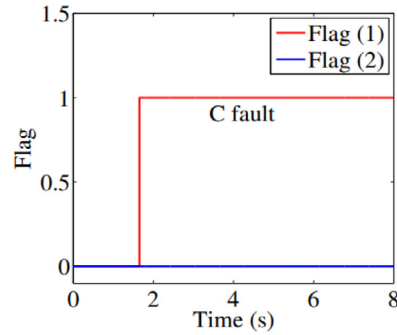
(a) Estimated value of inductance L parameters



(b) Estimated value of capacitance C parameters



(c) L and C residual diagram



(d) Fault detection flag

Fig. 3. Simulation of capacitance C fault.

4.1. Simulation of inductance L fault

Consider the case where the inductance L of the Boost converter circuit fails. The inductance C decreases from 5 mH to 3.5 mH in 1.5 s. The parameter estimation result of the inductance L is shown in Fig. 2(a). The estimated inductance L changes at 1.5 s. The tracking of the inductance L is shown in Fig. 2(b). Fig. 2(c) shows the relationship between the residual error of the inductance and capacitance and the threshold. According to the fault detection logic, the system detects the fault. The change diagram of the fault detection flag is shown in Fig. 2(d). At 1.5 s, the fault is detected.

4.2. Simulation of capacitance C fault

Considering the capacitor C failure of the boost converter, the capacitor changes from 2.85 mF to 1 mF in 1.5 s. The tracking of the inductance L is shown in Fig. 3(a). The parameter estimation result of the capacitance C is shown in Fig. 3(b). The relationship between the inductance residual and the capacitance residual and the threshold is shown in Fig. 3(c). It can be clearly observed that when the capacitor fails, the inductance residual is less than the threshold, and the capacitor residual exceeds the threshold. According to the

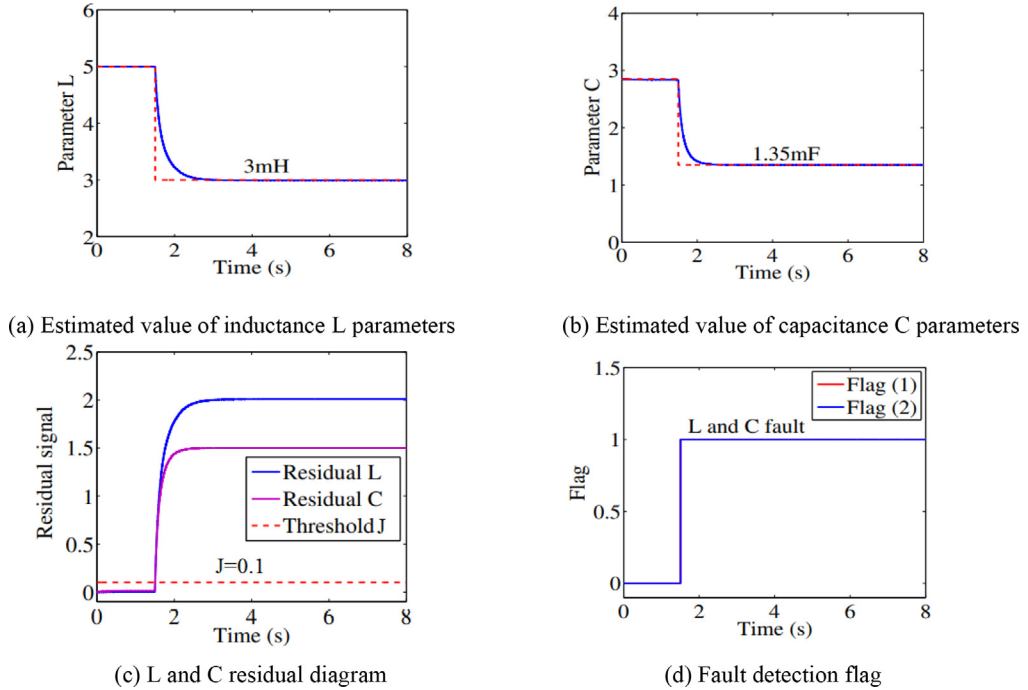


Fig. 4. Simulation of inductance L and capacitance C fault.

fault detection logic, it is judged that the capacitor is faulty. Fig. 3(d) describes the change curve of the fault detection flag.

4.3. Simulation of inductance L and capacitance C fault

In this subsection, we consider the case that the inductance L and capacitance C all occur faults in the Boost converter. The inductance changes from 5 mH to 3 mH in 1.5 s, while the capacitance C drops from 2.85 mF to 1.35 mF. The parameter estimation results of inductance L and capacitance C are shown in Fig. 4(a) and (b), separately. It can be seen from Fig. 4(c) that at 1.5 s, the residual error of the inductance L and capacitance C exceeds the threshold. At the same time, and it is judged that the inductance L and capacitance C are faulty at the same time. The change of the fault detection flag is shown in Fig. 4(d), and Flag (1) and Flag (2) overlap.

4.4. Simulation of inductance L fault with different noises

When the inductor is faulty. It can be seen from the figure that the inductance decreases from 5 mH to 1.5 mH at 0.1 s, and the fault damage degree is 70%, while the capacitance remains at 2.85 mF, almost unchanged. When the inductance and capacitance are at 1, 2 and 3 times of noise, the change values of inductance and capacitance estimated by the observer almost coincide completely, indicating that the adaptive sliding mode observer designed in this chapter has strong robustness, strong anti-interference ability to noise, and can accurately

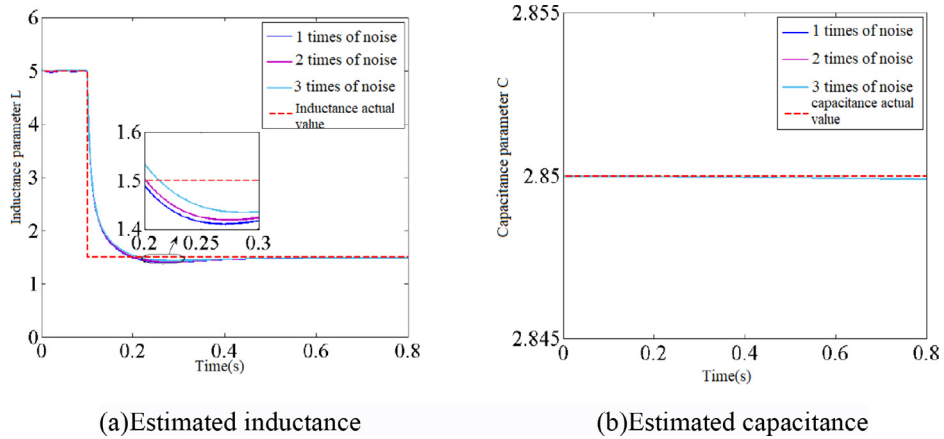
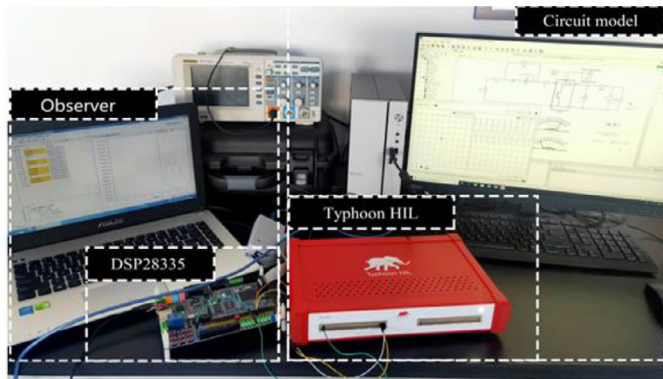
Fig. 5. Simulation of inductance L fault with different noises.

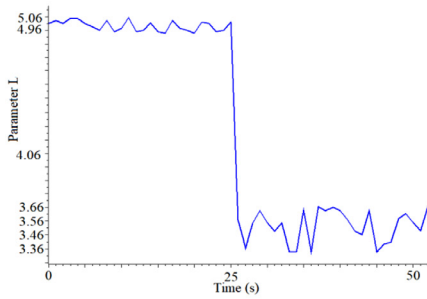
Fig. 6. Semi-physical simulation platform.

estimate the parameter values of components. Similarly, the observer can accurately estimate the inductance and capacitance of the Boost circuit under different noise levels (Fig. 5).

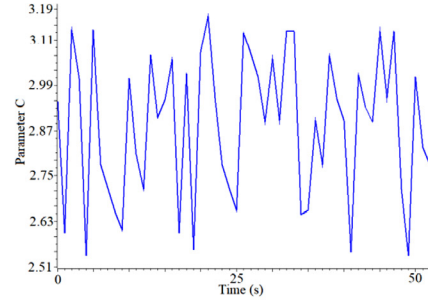
Remark 4. Comparing with the method in reference [18], the fault of inductor and capacitor is considered under the condition of adding noise. Simulation results show that the designed observer can accurately estimate the value of inductance and capacitance. Comparing with the method in reference [18], the adaptive sliding mode observer selected in this paper can reflect the value of inductance and capacitance faster and better when the fault occurs. With the addition of noise, the device parameters can be accurately estimated, and the fault detection logic can accurately locate the fault, which verifies the effectiveness of the proposed method.

5. Hardware-in-the-loop simulation

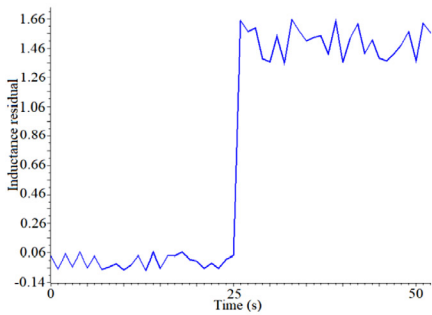
This section introduces the hardware-in-the-loop simulation verification of the fault detection algorithm. The hardware-in-the-loop simulator Typhoon-HIL402 and the digital controller DSP28335 constitute a hardware-in-the loop simulation platform. The boost circuit is built-in



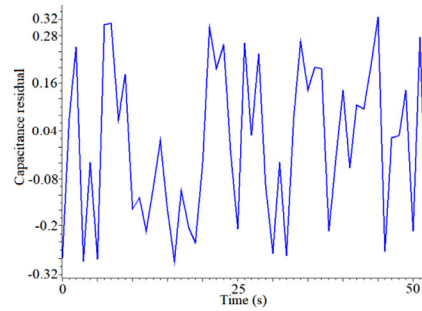
(a) Estimated value of inductance L parameters



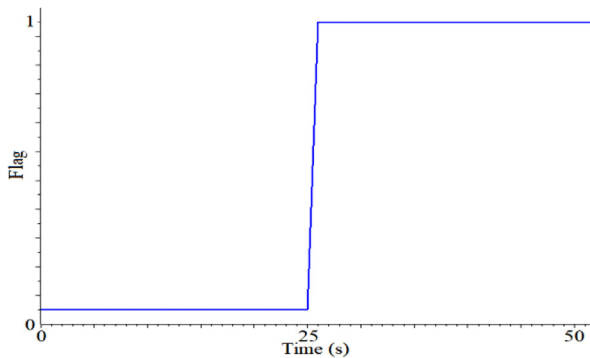
(b) Estimated value of capacitance C parameters



(c) residual r_L



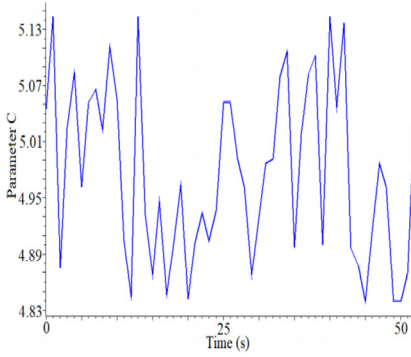
(d) residual r_C



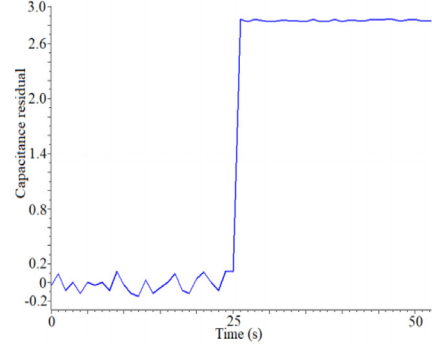
(e) Fault detection flag

Fig. 7. (a),(b) are the simulation results of the parameter estimation of the inductance L and capacitance C . (c), (d), (e) are simulation results for fault detection as inductance L occurs fault.

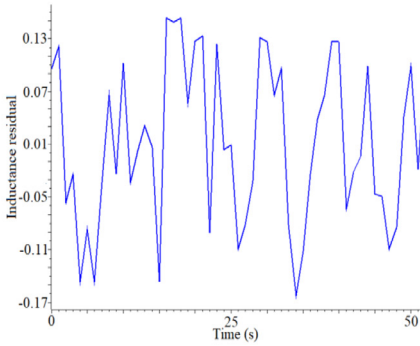
Typhoon-HIL402, and the fault detection algorithm is loaded in the DSP28335 in the form of C language. The structure of the semi-physical simulation system is indicated in Fig. 6. We simulated three types of faults in the typhoon to verify the effectiveness of the fault detection algorithm. The following part shows the simulation results of the three faults.



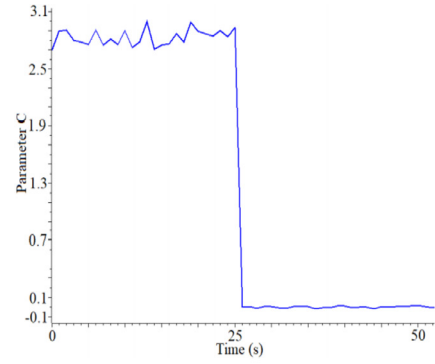
(a) Estimated value of inductance L parameters



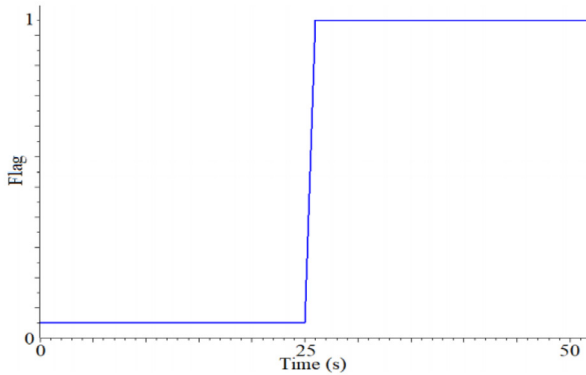
(b) Estimated value of capacitance C parameters



(c) residual r_L



(d) residual r_C

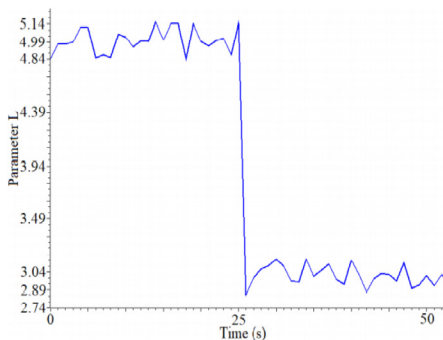


(e) Fault detection flag

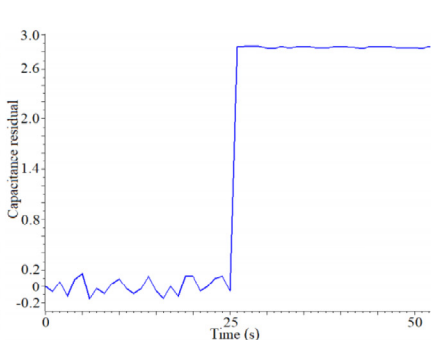
Fig. 8. (a),(b) are the simulation results of the parameter estimation of the inductance L and capacitance C . (c), (d), (e) are simulation results for fault detection as inductance C occurs fault.

5.1. Simulation of inductance L fault

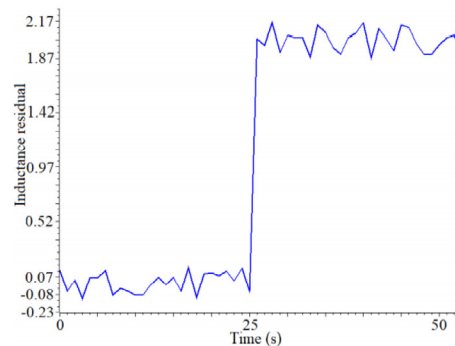
Consider the case where the inductance L of the Boost converter circuit fails. At 25 s, the inductance L dropped from 5 mH to 3.5 mH. The estimated figure of the inductance L parameter is shown as in Fig. 7(a). Fig. 7(b) shows the estimated diagram of the capacitance



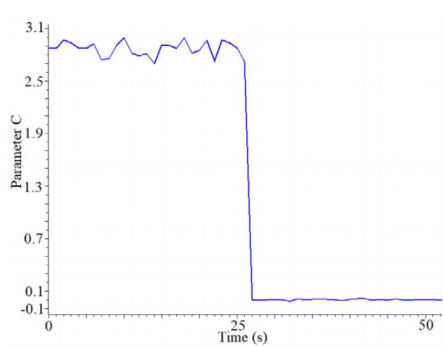
(a) Estimated value of inductance L parameters



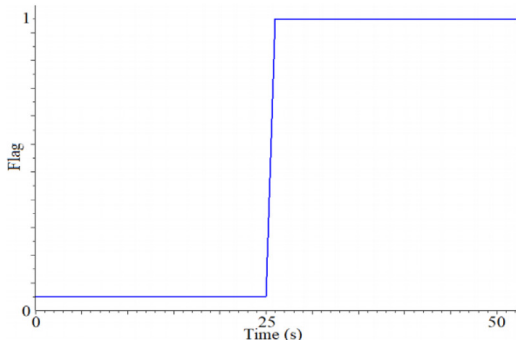
(b) Estimated value of capacitance C parameters



(c) residual r_L



(d) residual r_C



(e) Fault detection flag

Fig. 9. (a),(b) are the simulation results of the parameter estimation of the inductance L and capacitance C . (c), (d), (e) are simulation results for fault detection as inductance L and capacitance C occurs fault.

C after a failure. Fig. 7(c) and (d) represent the fault detection residuals of inductance L and capacitance C respectively. It can be seen from the Fig. 7(c) and (d) that when the L fault occurs, the residual value of the inductance L suddenly increases, and the residual value of the capacitors C very small and remains unchanged. According to the fault detection logic, the fault detection flag changes from 0 to 1 as shown in Fig. 7(e). The threshold is selected as 1.

5.2. Simulation of capacitance C fault

This section gives a half-physical simulation diagram of capacitor C failure. In the hardware-in-the-loop simulation, for better simulation effect, the capacitance C is reduced from 2.85 mF to 0.05 mF. Fig. 8(a) depicts the parameter estimation diagram of the inductance L value. Fig. 8(b) clearly shows the estimated capacitance C . The residual error diagrams of inductance L and capacitance C are shown in Fig. 8(c) and (d). At 25 s, the residual error of the inductance L is very small, and the residual value of the capacitance C becomes very large. According to the fault detection logic, the change diagram of the fault detection flag in Fig. 8(e) is obtained. The threshold is selected as 1.

5.3. Simulation of inductance L and capacitance C fault

When inductance L and capacitance C fail at the same time, the parameter estimation diagrams of inductance L and capacitance C are shown in Fig. 9(a) and (b). When the circuit fails, it can be seen from Fig. 9(c) and (d) that the residual signals of the inductance L and capacitance C have changed at the same time and exceeded the threshold, and the system has detected the fault. The threshold is selected as 1.

6. Conclusion

This paper proposes a DC-DC converter fault detection method based on adaptive parameter recognition. This method estimates component parameters in real-time via a new adaptive sliding mode observer which forms a residual signal through the estimated value and the actual value. Compared with the designed threshold, when the residual signal exceeds the threshold, the component fails. Simulations and semi-physical simulations were conducted using the boost circuit as an example. The simulation results show that the design of the adaptive sliding mode observer can quickly track component parameters and realize fault detection and identification through fault detection logic. The hardware-in-the-loop simulation verified the effectiveness of the method.

Declaration of Competing Interest

None.

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